Polymorphism Exercise

Create a C# Console App that demonstrates how different vehicles can be handled through one common base class.

1. Create a base class Vehicle with a public virtual void Move() method that prints "Vehicle moves".

2. Create Car and Bicycle classes that inherit from Vehicle. Override Move() so Car prints "Car drives" and Bicycle prints "Bicycle pedals".

3. Add a public void Honk() method to Car that prints "Beep!".

4. In Main, create a Vehicle[] containing one Car and one Bicycle. Use a foreach loop to call Move() on each element.

5. Use the is operator to check whether an element is a Car. If so, call Honk() safely. Add a short comment explaining why vehicle.Honk() cannot be called directly on a Vehicle variable.

Starter structure

```
using System;

class Vehicle
{
    // Write the virtual method.
}

class Car : Vehicle
{
    // Override Move() and add Honk().
}

class Bicycle : Vehicle
{
    // Override Move().
}

class Program
{
    static void Main()
    {
        // Create the array, loop through it, and test for Car.
    }
}
```

Think about

Why can a Vehicle variable refer to a Car object?

Why does Move() call the correct implementation?